Part I: Environment Setup

Option 1

Before beginning the assignment, create a dedicated Python environment for Deep Learning using **Anaconda Prompt**.

Step 1: Open Anaconda Prompt

Launch **Anaconda Navigator** and open **Anaconda Prompt**.

Step 2: Create a New Conda Environment

Create a new environment named **DeepLearning** with Python 3.11 (or the version recommended by the instructor).

```
conda create -n DeepLearning python=3.11
```

Step 3: Activate the Environment

```
conda activate DeepLearning
```

Verify the Python version.

```
python --version
```

Step 4: Install the Required Packages

Install the required libraries using pip.

```
pip install tensorflow ipykernel numpy pandas matplotlib scikit-learn opencv-python pillow torch torchvision
```

Step 5: Register the Environment as a Jupyter Kernel

```
python -m ipykernel install --user --name=DeepLearning --display-name="Python (Deep Learning)"
```

Step 6: Verify the Installation

Verify that the major libraries have been installed successfully.

```
pip list
```

Step 7: Launch Jupyter Notebook

Start Jupyter Notebook from the activated environment.

```
jupyter notebook
```

Step 8: Create the Notebook

1. From the Jupyter Notebook home page, click **New → Python (Deep Learning)**.
2. Create a new notebook.
3. Save it as: `ways_to_load_Images.ipynb`

Step 9: Verify the Environment Inside Jupyter Notebook

In the first cell of the notebook, execute the following commands to verify that the environment has been configured correctly.

```
import tensorflow as tf
import torch
import cv2
import PIL
import numpy as np
import matplotlib

print("TensorFlow Version :", tf.__version__)
print("PyTorch Version    :", torch.__version__)
print("OpenCV Version     :", cv2.__version__)
print("Pillow Version     :", PIL.__version__)
print("NumPy Version      :", np.__version__)
print("Matplotlib Version :", matplotlib.__version__)
```

Option 2: Create a separate environment and use it in Jupyter

If you want each project to have its own packages:

1. Create a virtual environment from the terminal:

```
python -m venv dl_env
```

2. Activate it and install packages:

```
pip install tensorflow ipykernel numpy pandas matplotlib scikit-learn opencv-
python pillow torch torchvision
```

3. Register it as a Jupyter kernel:

```
python -m ipykernel install --user --name=dl_env --display-name="Python (Deep Learning)"
```

4. Open Jupyter Notebook.

5. Go to **Kernel → Change Kernel → Python (Deep Learning)**.

Now your notebook runs inside the isolated environment.